Reyes, MARK JEROME G. May 29, 2024
3IT-C ITP 101 - Class Participation

I. Correlation Analysis

$n = 5$ $\Sigma x^2 = 220$
$\Sigma x = 30$ $\Sigma y^2 = 30,075$
$\Sigma y = 385$ $\Sigma xy = 2,440$

Question 1: Compute the Person Correlation Coefficient (r)

$$r = \frac{N\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{(N(\Sigma x^2)) - (\Sigma x)^2][N(\Sigma y^2) - (\Sigma y)^2]}}$$

$$r = \frac{5(2440) - (30)(385)}{\sqrt{(5(220) - 30^2)(5(30075) - 385^2)}} = \frac{12,200 - 11,550}{\sqrt{200 \times 2,150}} = \sqrt{430,000}$$

$$r = \frac{650}{655.74}$$ $r = 0.991$

Question 2: Interpret the Relationship

There is a very strong positive correlation between study hours and exam scores. As study hours increase, exam scores also increase consistently.

Question 3: Test of Significance ($\alpha = 0.05$)

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$t = \frac{0.991\sqrt{3-2}}{\sqrt{1-.9^2}} = \frac{0.991 \times \sqrt{3}}{0.134} = \frac{0.991 \times 1.732}{0.134}$$

$t = 12.81$

$df = 5 - 2 = 3$ (two-tailed, $\alpha = 0.05$, $df = 3$) $= 3.182$

II. Regression Analysis

$n=5 \quad \bar{x}=7 \quad \bar{y}=60 \quad x=35 \quad y=300 \quad x^2=255 \quad xy=2150$

Q1: Find the Regression Equation $y = a+bx$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = \frac{5(2150) - (35)(300)}{5(255) - 35^2} = \frac{10,750 - 10,500}{1275 - 1225}$$

$$b = \frac{250}{50} \qquad \boxed{b = 5}$$

$$a = \bar{y} - b\bar{x} = 60 - 5(7) = 60 - 35 \qquad \boxed{a = 25}$$

$$\boxed{\hat{y} = 25 + 5x}$$

Q2: Predict Sales Revenue when $x = 12$

$$\hat{y} = 25 + 5(12) = 25 + 60 = \boxed{\hat{y} = 85 \text{ (in thousands)}}$$

Q3: Interpretation the slope

The slope $b = 5$ means that for every ₱1,000 increase in advertising expense, sales revenue increase by ₱5,000. There is a direct and proportional positive relationship between advertising and sales.

III. One-way ANOVA

Q1: state the Hypotheses

- H_0: $\mu A = \mu B = \mu C$ – there is no significant difference in test scores among the three teaching methods.
- H_1: At least one group mean is significantly different from the others

Computations

	Method A	Method B	Method C
Score1	78	85	40
Score2	80	99	42
Scor 3	82	96	91
Sum (T_j)	240	259	273
Mean	80	86.33	91

$n = 9$ Grand total $T = 240 + 259 + 273 = 772$ ~~Grand mean~~

Grand mean $= 772/9 = 85.78$

$$CF = \frac{T^2}{n} = \frac{772^2}{9} = \frac{596,\ 214}{9} = 66{,}253.78$$

$$SS\ Total = (78^2 + 90^2 + 82^2 + 85^2 + 99^2 + 96^2 + 40^2 + 42^2 + 91^2) - CF$$

$$= 6084 + 6400 + 6724 + 7225 + 7744 + 7396 + 8100 + 8464 + 8281) - 66{,}253.78$$

$$= 66{,}418 - 66{,}253.78$$

$$\boxed{SS\ Total = 164.22}$$

$$SS\ Between = \Sigma(T_j/n_j) - CF = (240^2/3 + 259^2/3 + 273^2/3) - 66{,}253.78$$

$$= (19{,}200 + 22{,}374.33 + 24{,}843) - 66{,}253.78$$

$$= 66{,}417.33 - 66{,}253.78$$

$$\boxed{SS\ Between = 163.56}$$

$$SS\ within = SS\ total - SS\ Between = 164.22 - 163.56$$

$$\boxed{SS\ within = 0.67}$$

IV. Two-Way ANOVA

Q1. Test - Significant Difference Among Teaching Methods

H_0: There is no significant difference in student performance among the three teaching methods.

H_1: At least one teaching method produces significantly different performance.

Q2. Test - Significant Difference Among Subjects

H_0: There is no significant difference in student performance among the three subject areas.

H_1: At least one subject area shows significantly different student performance.

Data & Computation

Subject	Method A	Method B	Method C	Row total
Math	85	88	90	263
English	80	84	86	250
Science	82	85	89	256
Col Total	247	257	265	T = 769

$N = 9$ $r = 3$ rows $c = 3$ columns

$CF = T^2/N = 769^2/9 = 591,361/9 = \underline{65,706.78}$

SS Total $= \sum y^2 - CF = (85^2 + 88^2 + 90^2 + 80^2 + 84^2 + 86^2 + 82^2 + 85^2 + 89^2) - CF$

$= (7225 + 7744 + 8100 + 6400 + 7056 + 7396 + 6724 + 7225 + 7921) - 65,706.78$

$= 65,791 - 65,706.78$

SS total = 84.22

SS columns (Methods) $= \sum (C_j^2/r) - CF = (247^2/3 + 257^2/3 + 265^2/3) - CF$

$= 20,336.33 + 22,003.33 + 23,401.33) - 65,706.78$

SS Methods = 41.22

SS Rows (Subjects) $= \sum(R_i^2/c) - CF = (263^2/3 + 250^2/3 + 256^2/3) - CF$

$= (23,036.33 + 20,833.33 + 21,845.33) - 65,706.78$

$= 65,735 - 65,706.78$

SS Subjects = 28.22

SS Error = SS Total − SS Methods − SS Subjects $= 84.22 - 41.22 - 28.22$

SS Error = 14.78

Q3: Complete the ANOVA table

Source of Variation	SS	df	MS	F
Methods (columns)	41.22	2	20.61	11.17
Subjects (Rows)	28.22	2	14.11	7.65
Error (Interaction)	14.78	4	1.85	—
Total	84.22	8	—	—

dF Methods $= c - 1 = 2$ dF Error $= (r-1)(c-1) = 4$

dF Subjects $= r - 1 = 2$ F critical $(dF1 = 2, dF2 = 4, \alpha = 0.05) = \underline{6.94}$

Q4: Interpret the Result

Teaching Methods: F = 11.17 > F critical 6.94 → Reject H_0. There is a significant difference in performance due to teaching method.

Subject Areas: F = 7.65 > F-critical 6.94 → Reject H_0. There is a significant difference in performance across subject areas. Both factors (teaching method and subjects) significantly affect student performance.